

Mukul Rayana

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TECHNICAL SKILLS

Languages: Python, SQL, C++, Bash

ML / DL: PyTorch, HuggingFace Transformers, Scikit-learn, XGBoost, ONNX, NumPy, Pandas

GenAI / Agents: LangGraph, LangChain, Llama, QLoRA, PEFT, Groq, FAISS, Pinecone, ChromaDB

MLOps / Infra: FastAPI, Docker, MLflow, GitHub Actions, AWS (EC2, S3), Azure Container Apps

AI Safety / Eval: DeBERTa NLI, NeMo Guardrails, LLM-as-a-Judge, SHAP, Captum

EDUCATION

University of Maryland, College Park

M.S., Applied Machine Learning

• **GPA:** 3.5 / 4.0

Aug 2024 - May 2026

College Park, MD

SRM Easwari Engineering College

B.Tech., AI and Data Science

• **GPA:** 3.55 / 4.0

Dec 2020 - Apr 2024

Chennai, India

EXPERIENCE

BeeBox Studios Pvt Ltd | IIT Madras Research Park

AI Research Intern

Jun 2023 - Apr 2024

Chennai, India

- Shipped a BERT-based QA system for real-time in-headset queries over a 100-page XR corpus, hitting 83% F1; ONNX export and INT8 quantization cut P95 latency from 185 ms to 62 ms on GPU-constrained hardware.
- Owned the ML deployment pipeline end-to-end (Docker, GitHub Actions CI/CD, zero-downtime AWS EC2), serving as the sole ML point of contact for UI/UX and WebRTC teams, which kept the XR prototype on schedule and enabled continuous model updates without service interruption

PROJECTS

PharmaChain [Live Demo](#)

Apr 2026

- LangGraph, XGBoost, FastAPI, pgvector, Supabase, React, Optuna, SHAP
- 1st Place, UMD Smith Agentic AI Challenge (\$4K team prize)
- Collaborated with a team of 5; built the Route, Triage, and Insurance agents within an 8-agent LangGraph orchestrator that resolves cold-chain incidents in 12–15 seconds, with a human-approval gate blocking all irreversible actions.
- Combined 8 deterministic rules with an Optuna-tuned XGBoost classifier (0.94 ROC-AUC, 7,411 telemetry windows) and SHAP explanations; RAG over 417 WHO/FDA/EU GDP regulatory chunks grounds every compliance decision.

EmpathRAG [Live Demo](#)

Feb 2026 - Apr 2026

- DeBERTa, RoBERTa, LoRA, FAISS, Groq, Captum, Gradio
- Separated all safety decisions from generation: a deterministic planner owns routing; the LLM paraphrases only; a post-generation verifier rejects scope drift – 0/28 missed escalations vs. 9/28 (32.1%) for the unguarded baseline.
- Fine-tuned DeBERTa-v3 on 230K NLI pairs for crisis detection; 96.3% recall and 100% recall on euphemistic phrasing vs. 20% for keyword filters; grounded all responses in a curated registry of 34 verified resources.

Nemesis [GitHub](#)

Nov 2025 - Mar 2026

- PPO, Unity ML-Agents, Ray RLlib, ONNX, Unity Sentis, Groq, NeMo Guardrails
- Trained a PPO policy over 300K steps with a 3-phase curriculum (reward -2.4 to 14.93); RL boss outlasts scripted FSM by 85% (37 s vs. 20 s, 50 episodes); ONNX model runs via Unity Sentis at under 2 ms/frame, 60 fps.
- Scaled distributed training to a 2.62x speedup (77.2 s to 29.5 s, 1 to 8 Ray RLlib workers); integrated a Groq + NeMo Guardrails TCP bridge generating filtered in-character dialogue every 6 seconds.

Irminsul [Live Demo](#)

Nov 2025 - Jan 2026

- QLoRA, PEFT, Pinecone, FastAPI, MLflow, Docker, Azure Container Apps
- Fine-tuned Llama 3.1 8B with QLoRA across 3 MLflow runs (faithfulness 0.826, ROUGE-L 0.466); deployed via Dockerized FastAPI on Azure Container Apps with a 6,876-vector Pinecone index; automated weekly corpus refresh via GitHub Actions.

PUBLICATION

- Voice-Driven Panoramic VR Generation. *IEEE IDCIoT 2024 (peer-reviewed, co-author) – speech-to-360° pipeline; 2.3× throughput via attention slicing (Whisper → Stable Diffusion).*